

# Distributed Microservices Bases E-Commerce Platform

## Database Schema

There are total 3 tables

### 1. Users

| Column Name | Data Type    | Constraints                               | Description                                         |
|-------------|--------------|-------------------------------------------|-----------------------------------------------------|
| user_id     | INT          | PRIMARY KEY,<br>AUTO_INCREMENT            | Unique ID for each user                             |
| username    | VARCHAR(50)  | UNIQUE, NOT NULL                          | Login username                                      |
| password    | VARCHAR(255) | NOT NULL                                  | Encrypted password                                  |
| email       | VARCHAR(100) | UNIQUE, NOT NULL                          | User email                                          |
| role        | ENUM         | ('customer','vendor','admin')<br>NOT NULL | Defines user type                                   |
| phone       | VARCHAR(10)  | UNIQUE ,NOT NULL                          | Customer/vendor phone number                        |
| address     | VARCHAR(255) | NOT NULL                                  | Customer shipping address / Vendor business address |
| created_at  | TIMESTAMP    | DEFAULT<br>CURRENT_TIMESTAMP              | Account creation timestamp                          |

### 2. Products

| Column Name | Data Type     | Data Type                       | Description                  |
|-------------|---------------|---------------------------------|------------------------------|
| product_id  | INT           | PRIMARY KEY,<br>AUTO_INCREMENT  | Unique ID for each product   |
| name        | VARCHAR(100)  | NOT NULL                        | Product name                 |
| description | TEXT          | NOT NULL                        | Product description          |
| price       | DECIMAL(10,2) | NOT NULL                        | Product price                |
| vendor_id   | INT           | FOREIGN KEY →<br>users(user_id) | Vendor who added the product |
| created_at  | TIMESTAMP     | DEFAULT<br>CURRENT_TIMESTAMP    | Product creation timestamp   |

### 3. Orders

| Column Name | Data Type | Data Type                                                                   | Description                 |
|-------------|-----------|-----------------------------------------------------------------------------|-----------------------------|
| order_id    | INT       | PRIMARY KEY, AUTO_INCREMENT                                                 | Unique ID for each order    |
| customer_id | INT       | FOREIGN KEY → users(user_id)                                                | Customer placing the order  |
| product_id  | INT       | FOREIGN KEY → products(product_id)                                          | Ordered product             |
| vendor_id   | INT       | FOREIGN KEY → users(user_id)                                                | Vendor fulfilling the order |
| status      | ENUM      | ('placed','cancelled','shipped','delivered','returned')<br>DEFAULT 'placed' | Current order status        |
| order_date  | TIMESTAMP | DEFAULT CURRENT_TIMESTAMP                                                   | Order placement date        |

### Relationships

- Users → Products: One vendor can add many products (**One-to-Many**).
- Users → Orders: One customer can place many orders (**One-to-Many**).
- Products → Orders: One product can appear in many orders (**One-to-Many**).